

✓ TEXT TO IMAGE

✓ Adding your Hugging Face Token to Colab Secrets

1. **Get your HF Token:** Go to [Hugging Face settings](#) and generate a new access token (if you don't have one).
2. **Open Colab Secrets:** In your Colab notebook, click on the "🔑" (key) icon in the left sidebar.
3. **Add Secret:** Click 'Add new secret'.
4. **Name the Secret:** For the 'Name' field, type `HF_TOKEN` (this is case-sensitive and important).
5. **Paste your Token:** For the 'Value' field, paste your Hugging Face token.
6. **Save:** Ensure 'Notebook access' is enabled for this secret.

Once added, you can access the secret in your code like this:

```
!nvidia-smi
```

```
Sat Jun 20 13:10:13 2026
```

```
+-----+
| NVIDIA-SMI 580.82.07                  Driver Version: 580.82.07          CUDA Versi
+-----+-----+-----+-----+-----+
| GPU   Name                Persistence-M | Bus-Id        Disp.A | Volatile
| Fan   Temp   Perf          Pwr:Usage/Cap |      Memory-Usage | GPU-Util
+-----+-----+-----+-----+-----+
|  0    Tesla T4               Off          | 00000000:00:04.0 Off |
| N/A   43C    P8             10W / 70W   |  0MiB / 15360MiB |      0%
+-----+-----+-----+-----+-----+
```

```
+-----+
| Processes:
```

```
| GPU   GI    CI          PID    Type    Process name
|      ID    ID
+-----+
| No running processes found
+-----+
```

```
!pip install diffuser --upgrade
```

```
Collecting diffuser
```

```
  Downloading diffuser-0.0.1.tar.gz (1.6 kB)
```

```
  Preparing metadata (setup.py) ... done
```

```
Building wheels for collected packages: diffuser
```

```
  Building wheel for diffuser (setup.py) ... done
```

```
  Created wheel for diffuser: filename=diffuser-0.0.1-py3-none-any.whl size=1
```

```

Stored in directory: /root/.cache/pip/wheels/9d/33/60/eaac3227d4a649668ab6c
Successfully built diffuser
Installing collected packages: diffuser
Successfully installed diffuser-0.0.1

```

```
!pip install invisible_watermark transformers accelerate safetensors
```

```

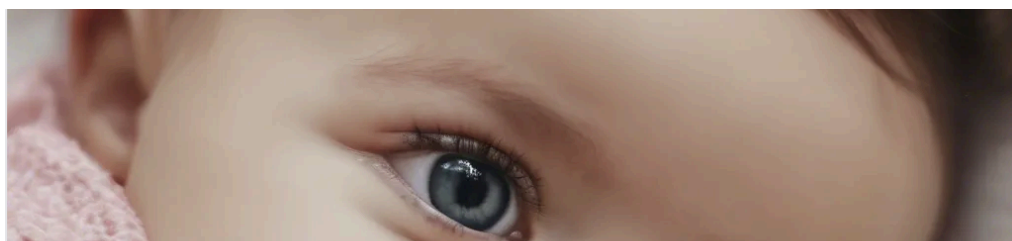
Requirement already satisfied: PyWavelets>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: numpy>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: opencv-python>=4.1.0.25 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers==4.46.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers==4.46.2)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers==4.46.2)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from packaging>=20.0->transformers==4.46.2)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers==4.46.2)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from invisible_watermark==0.0.1)

```

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12,
 Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist
 Downloading invisible_watermark-0.2.0-py3-none-any.whl (1.6 MB)
 1.6/1.6 MB 31.3 MB/s eta 0:00:00
 Installing collected packages: invisible_watermark

```
import torch
from diffusers import DiffusionPipeline
import gradio as gr
# Load Stable Diffusion XL model
pipe = DiffusionPipeline.from_pretrained(
    "stabilityai/stable-diffusion-xl-base-1.0",
    torch_dtype=torch.float16,
    use_safetensors=True,
    variant="fp16"
)
pipe.to("cuda")
# Image generation function
def generate_image(prompt):
    image = pipe(prompt=prompt).images[0]
    return image
# Gradio Interface
demo = gr.Interface(
    fn=generate_image,
    inputs=gr.Textbox(label="Enter your prompt"),
    outputs=gr.Image(label="Generated Image"),
    title="Stable Diffusion XL Image Generator",
    description="Generate AI images using Stable Diffusion XL"
)
demo.launch()
```

Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recom
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recom
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: U
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings ta
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access p
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
model_index.json: 100% 609/609 [00:00<00:00, 54.7kB/s]
Download complete: 100% 7.11G/7.11G [01:08<00:00, 91.0MB/s]
Fetching 19 files: 100% 19/19 [01:05<00:00, 4.31s/it]
Loading pipeline components...: 100% 7/7 [00:04<00:00, 1.44it/s]
Loading weights: 100% 196/196 [00:00<00:00, 757.56it/s]
Loading weights: 100% 517/517 [00:00<00:00, 691.82it/s]
It looks like you are running Gradio on a hosted Jupyter notebook, which requ
Colab notebook detected. To show errors in colab notebook, set debug=True in
* Running on public URL: <https://b4ff28ca1fc2732db5.gradio.live>
This share link expires in 1 week. For free permanent hosting and GPU upgrade



Next steps:  Deploy to Cloud Run

Start coding or generate with AI.